

A Subject Project Proposal Report on

Multiplayer Tic-Tac-Toe over TCP

Submitted in Partial Fulfilment of the Requirements for
the Degree of **Software Engineering**
under Pokhara University

Submitted by:

Dikshit Neupane, 231613
Prem Raj Chaudhary, 231631
Success K.C., 231646

Under the supervision of

Madan Kadariya

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Department of Software Engineering
NEPAL COLLEGE OF INFORMATION TECHNOLOGY
Balkumari, Lalitpur, Nepal

1. Introduction:

Games based on network can be used as an efficient medium for learning about client-server communication and socket programming. The current project aims at building a Multiplayer Tic Tac Toe game by using C++ programming language and the Winsock API, which works over the TCP protocol. The game enables two people to play the game in real time by connecting to the main server over a local network.

The main responsibility of the server includes taking care of player connections, checking move validity, maintaining game state and updating the clients with information. Thus, the users get practical knowledge in TCP socket programming, data communication and real-time communication via a network.

2. Objectives:

- Design a client/server Tic-Tac-Toe game using Winsock.
- Use TCP sockets for effective communication.
- Coordinate turn-by-turn play among two clients.
- Check validity of move and decide the result of the game.
- Enhance practical skills about socket programming and networking.

3. Methodology:

The development of the project will be conducted based on the client-server architecture, the TCP protocol, and the Winsock API in C++. At first, the server program will be developed to accept the connections from the clients and connect two users. Then, the server will assign the symbols X and O and launch the game.

In the course of the game, every client will send its move to the server. The server will validate the move, update the game field, check whether there is a winner or the match has ended in a draw, and send the updated information to both clients. In addition, the server will make sure that all players make their move according to the rules. After the end of the match, the players will be able either to restart the game or to exit the application. The testing of the project will be performed on a local network.

4. Expected Outcome:

- Working version of multiplayer Tic-Tac-Toe over a local network.
- Effective communication between two clients via a server.
- Real-time updating of the game along with proper management of turns.
- Detection of win, loss, and draw.
- The system will be tested on a local network to check for effective communication, proper game synchronization, and other functionalities.

5. Technology Used:

- Programming Language: C++
- Networking API: Winsock2
- Protocol: TCP
- Platform: Windows
- IDE: Visual Studio

6. Conclusion:

This project gives hands-on practice of TCP socket programming using Winsock in building a real time multiplayer game. It is a good example of how client/server communication and synchronization can be applied through network programming.